

Project Initialization and Planning Phase

Date	02-July-2024
Team ID	XXXXXXX
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to replace traditional manual credit screening barriers by developing an automated underwriting system powered by a Random Forest Classifier to ensure exceptionally fast, data-driven loan eligibility predictions.
Scope	The project encompasses building a comprehensive data preprocessing, balancing, and standardization pipeline linked directly to a responsive multi-column browser console to yield real-time risk evaluations.
Problem Statement	
Description	Inefficiencies and human variance in traditional manual credit scoring systems cause massive operational delays. Furthermore, uncompensated class imbalances within historical financial records degrade algorithmic evaluation neutrality, creating artificial boundaries for viable retail loan applicants.
Impact	Automating this framework minimizes operational processing fatigue for bank credit officers, bypasses missing data halts through strict input validation enforcements, reduces default assets, and significantly optimizes customer transaction confidence.
Proposed Solution	
Approach	Deploying a localized, multi-threaded Flask web architecture unpickling pre-fit machine learning vectors to provide uniform standard score feature normalization and instant random forest tree splits on incoming financial payloads.
Key Features	- Data Equity Calibration: Minority neighborhood oversampling via SMOTE to balance targets and remove dataset skews. - Dynamic Standardization: Continuous feature magnitude scaling

	using pre-compiled StandardScaler objects (scale1.pkl). - Sub-0.2s Local Execution: Rapid server inference via unpickled ensemble structures (rdf.pkl) running without external API dependencies. - Contextual Jinja2 Interface: Viewports that adapt color states and typography layouts to dynamically match approval determinations.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Standard Multi-Threaded Processor Core Configuration	Intel Core i5 / AMD Ryzen 5 or equivalent (Minimum 4 Cores)
Memory	Local Execution RAM Buffer	8 GB
Storage	Workspace Flat SSD Disk Real Estate	256 GB NVMe SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, imbalanced-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook & Visual Studio Code (VS Code)
Data		
Data	Comprehensive Tabular Loan Prediction Dataset	Kaggle Financial Tabular Corpus (loan_prediction.csv), 614 Records, CSV Format